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DEVICES, METHODS, APPARATUSES, AND COMPUTER READABLE MEDIA FOR MEDIUM ACCESS CONTROL- CONTROL ELEMENT TRANSMISSION

Abstract

Disclosed are devices, methods, apparatuses, and computer readable media for medium access control-control element transmission. An example terminal device may include at least one processor and at least one memory. The at least one memory may include computer program code, and the at least one memory and the computer program code may be configured to, with the at least one processor, cause the terminal device to perform: obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element; and sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

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Background/Summary

TECHNICAL FIELD

[0001] Various embodiments relate to devices, methods, apparatuses, and computer readable media for medium access control-control element (MAC-CE) transmission.

BACKGROUND

[0002] In a non-terrestrial network (NTN), for uplink (UL), two UL hybrid automatic repeat request (HARQ) retransmission states/modes are defined: HARQ mode A and HARQ mode B. UE behavior in the HARQ mode A best supports reception of UL retransmission grant based on UL decoding result. For example, in the HARQ mode A, the HARQ retransmission may be scheduled by network side based on the HARQ transmission decoding result. UE behavior in the HARQ mode B best supports no UL retransmission and/or blind UL retransmission. For example, in the HARQ mode B, the HARQ retransmission may be scheduled before the network side achieving the HARQ transmission decoding result. When a MAC-CE is multiplexed/transmitted, the trigger event for the MAC-CE will be cancelled. If the MAC-CE is multiplexed in the HARQ mode B, no UL retransmission and/or blind UL retransmission cannot ensure the MAC-CE can be decoded correctly. If the MAC-CE is critical and cannot be decoded by a base station (BS) successfully, system performance will be impacted. On the other hand, if the MAC-CE is limited to be multiplexed in the HARQ mode A, the MAC-CE transmission opportunity is also limited, which may cause transmission latency.

SUMMARY

[0003] A brief summary of exemplary embodiments is provided below to provide basic understanding of some aspects of various embodiments. It should be noted that this summary is not intended to identify key features of essential elements or define scopes of the embodiments, and its sole purpose is to introduce some concepts in a simplified form as a preamble for a more detailed description provided below.

[0004] In a first aspect, disclosed is a terminal device. The terminal device may include at least one processor and at least one memory. The at least one memory may include computer program code, and the at least one memory and the computer program code may be configured to, with the at least one processor, cause the terminal device to perform: obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element; and sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0005] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the terminal device to further perform: cancelling the trigger event if the uplink medium access control-control element is sent in the hybrid automatic repeat request process of mode A; and refraining from further sending the uplink medium access control-control element.

[0006] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the terminal device to further perform: cancelling the trigger event if receiving, from the network device, a downlink medium access control-control element acknowledging decoding of the uplink medium access control-control element; and refraining from further sending the uplink medium access control-control element.

[0007] In some example embodiments, the acknowledging downlink medium access control-

control element may have a logical channel identifier or an extended logical channel identifier specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0008] In some example embodiments, the logical channel identifier or the extended logical channel identifier may be in a subheader of the acknowledging downlink medium access control-control element, and the acknowledging downlink medium access control-control element may be without a payload.

[0009] In some example embodiments, the downlink medium access control-control element may have a reserved bit specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0010] In some example embodiments, the acknowledging downlink medium access control-control element may have a payload indicating the acknowledgment of decoding the uplink medium access control-control element by a bitmap.

[0011] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the terminal device to further perform: cancelling the trigger event if the number of transmission reaches the configured transmission number.

[0012] In some example embodiments, the number of transmission may be counted based on a number of initial transmission and retransmission of at least one transport block carrying the uplink medium access control-control element.

[0013] In some example embodiments, the number of transmission may be counted based on a number of initial transmission of at least one transport block carrying the uplink medium access control-control element.

[0014] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the terminal device to further perform: starting a timer configured by the network device for retriggering the trigger event, and retriggering the trigger event upon expiry of the timer if the number of transmission reaches the configured transmission number and without receiving, from the network device, a downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0015] In some example embodiments, the timer may be started at a last retransmission of the uplink medium access control-control element.

[0016] In some example embodiments, the timer may be started at an initial transmission of the uplink medium access control-control element.

[0017] In some example embodiments, the uplink medium access control-control element may be sent in the hybrid automatic repeat request process of mode B.

[0018] In some example embodiments, the transmission number may be common for a plurality of uplink medium access control-control elements.

[0019] In some example embodiments, the transmission number may be specific to the uplink medium access control-control element.

[0020] In a second aspect, disclosed is a network device. The network device may include at least one processor and at least one memory. The at least one memory may include computer program code, and the at least one memory and the computer program code may be configured to, with the at least one processor, cause the network device to perform: configuring a transmission number for a terminal device to send an uplink medium access control-control element, wherein the uplink medium access control-control element may be sent to the network device in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0021] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the network device to further

perform: transmitting, to the terminal device, a downlink medium access control-control element for acknowledging decoding of the uplink medium access control-control element if decoding the uplink medium access control-control element successfully.

[0022] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the network device to further perform: configuring the downlink medium access control-control element.

[0023] In some example embodiments, the at least one memory and the computer program code may be further configured to, with the at least one processor, cause the network device to further perform: configuring a timer for the terminal device to retrigger the trigger event, the trigger event being retriggered upon expiry of the timer if the number of transmission reaches the transmission number and the terminal device without receiving, from the network device, the downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0024] In a third aspect, disclosed is a method performed by a terminal device. The method may comprise: obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element; and sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0025] In some example embodiments, the method may further comprise: cancelling the trigger event if the uplink medium access control-control element is sent in the hybrid automatic repeat request process of mode A; and refraining from further sending the uplink medium access control-control element.

[0026] In some example embodiments, the method may further comprise: cancelling the trigger event if receiving, from the network device, a downlink medium access control-control element acknowledging decoding of the uplink medium access control-control element; and refraining from further sending the uplink medium access control-control element.

[0027] In some example embodiments, the acknowledging downlink medium access control-control element may have a logical channel identifier or an extended logical channel identifier specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0028] In some example embodiments, the logical channel identifier or the extended logical channel identifier may be in a subheader of the acknowledging downlink medium access control-control element, and the acknowledging downlink medium access control-control element may be without a payload.

[0029] In some example embodiments, the downlink medium access control-control element may have a reserved bit specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0030] In some example embodiments, the acknowledging downlink medium access control-control element may have a payload indicating the acknowledgment of decoding the uplink medium access control-control element by a bitmap.

[0031] In some example embodiments, the method may further comprise: cancelling the trigger event if the number of transmission reaches the configured transmission number.

[0032] In some example embodiments, the number of transmission may be counted based on a number of initial transmission and retransmission of at least one transport block carrying the uplink medium access control-control element.

[0033] In some example embodiments, the number of transmission may be counted based on a number of initial transmission of at least one transport block carrying the uplink medium access control-control element.

[0034] In some example embodiments, the method may further comprise: starting a timer

configured by the network device for retriggering the trigger event, and retriggering the trigger event upon expiry of the timer if the number of transmission reaches the configured transmission number and without receiving, from the network device, a downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0035] In some example embodiments, the timer may be started at a last retransmission of the uplink medium access control-control element.

[0036] In some example embodiments, the timer may be started at an initial transmission of the uplink medium access control-control element.

[0037] In some example embodiments, the uplink medium access control-control element may be sent in the hybrid automatic repeat request process of mode B.

[0038] In some example embodiments, the transmission number may be common for a plurality of uplink medium access control-control elements.

[0039] In some example embodiments, the transmission number may be specific to the uplink medium access control-control element.

[0040] In a fourth aspect, disclosed is a method performed by a network device. The method may comprise: configuring a transmission number for a terminal device to send an uplink medium access control-control element, wherein the uplink medium access control-control element may be sent to the network device in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0041] In some example embodiments, the method may further comprise: transmitting, to the terminal device, a downlink medium access control-control element for acknowledging decoding of the uplink medium access control-control element if decoding the uplink medium access control-control element successfully.

[0042] In some example embodiments, the method may further comprise: configuring the downlink medium access control-control element.

[0043] In some example embodiments, the method may further comprise: configuring a timer for the terminal device to retrigger the trigger event, the trigger event being retriggered upon expiry of the timer if the number of transmission reaches the transmission number and the terminal device without receiving, from the network device, the downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0044] In a fifth aspect, disclosed is an apparatus. The apparatus as a terminal device may comprise: means for obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element; and means for sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0045] In some example embodiments, the apparatus may further comprise: means for cancelling the trigger event if the uplink medium access control-control element is sent in the hybrid automatic repeat request process of mode A; and means for refraining from further sending the uplink medium access control-control element.

[0046] In some example embodiments, the apparatus may further comprise: means for cancelling the trigger event if receiving, from the network device, a downlink medium access control-control element acknowledging decoding of the uplink medium access control-control element; and means for refraining from further sending the uplink medium access control-control element.

[0047] In some example embodiments, the acknowledging downlink medium access control-control element may have a logical channel identifier or an extended logical channel identifier specific for the acknowledgment of the decoding of the uplink medium access control-control

element.

[0048] In some example embodiments, the logical channel identifier or the extended logical channel identifier may be in a subheader of the acknowledging downlink medium access control-control element, and the acknowledging downlink medium access control-control element may be without a payload.

[0049] In some example embodiments, the downlink medium access control-control element may have a reserved bit specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0050] In some example embodiments, the acknowledging downlink medium access control-control element may have a payload indicating the acknowledgment of decoding the uplink medium access control-control element by a bitmap.

[0051] In some example embodiments, the apparatus may further comprise: means for cancelling the trigger event if the number of transmission reaches the configured transmission number.

[0052] In some example embodiments, the number of transmission may be counted based on a number of initial transmission and retransmission of at least one transport block carrying the uplink medium access control-control element.

[0053] In some example embodiments, the number of transmission may be counted based on a number of initial transmission of at least one transport block carrying the uplink medium access control-control element.

[0054] In some example embodiments, the apparatus may further comprise: means for starting a timer configured by the network device for retriggering the trigger event, and means for retriggering the trigger event upon expiry of the timer if the number of transmission reaches the configured transmission number and without receiving, from the network device, a downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0055] In some example embodiments, the timer may be started at a last retransmission of the uplink medium access control-control element.

[0056] In some example embodiments, the timer may be started at an initial transmission of the uplink medium access control-control element.

[0057] In some example embodiments, the uplink medium access control-control element may be sent in the hybrid automatic repeat request process of mode B.

[0058] In some example embodiments, the transmission number may be common for a plurality of uplink medium access control-control elements.

[0059] In some example embodiments, the transmission number may be specific to the uplink medium access control-control element.

[0060] In a sixth aspect, disclosed is an apparatus. The apparatus as a network device may comprise: means for configuring a transmission number for a terminal device to send an uplink medium access control-control element, wherein the uplink medium access control-control element may be sent to the network device in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0061] In some example embodiments, the apparatus may further comprise: means for transmitting, to the terminal device, a downlink medium access control-control element for acknowledging decoding of the uplink medium access control-control element if decoding the uplink medium access control-control element successfully.

[0062] In some example embodiments, the apparatus may further comprise: means for configuring the downlink medium access control-control element.

[0063] In some example embodiments, the apparatus may further comprise: means for configuring a timer for the terminal device to retrigger the trigger event, the trigger event being retriggered upon expiry of the timer if the number of transmission reaches the transmission number and the

terminal device without receiving, from the network device, the downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0064] In a seventh aspect, a computer readable medium is disclosed. The computer readable medium may include instructions stored thereon for causing a terminal device to perform: obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element: and sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0065] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the terminal device to further perform: cancelling the trigger event if the uplink medium access control-control element is sent in the hybrid automatic repeat request process of mode A: and refraining from further sending the uplink medium access control-control element.

[0066] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the terminal device to further perform: cancelling the trigger event if receiving, from the network device, a downlink medium access control-control element acknowledging decoding of the uplink medium access control-control element: and refraining from further sending the uplink medium access control-control element.

[0067] In some example embodiments, the acknowledging downlink medium access control-control element may have a logical channel identifier or an extended logical channel identifier specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0068] In some example embodiments, the logical channel identifier or the extended logical channel identifier may be in a subheader of the acknowledging downlink medium access control-control element, and the acknowledging downlink medium access control-control element may be without a payload.

[0069] In some example embodiments, the downlink medium access control-control element may have a reserved bit specific for the acknowledgment of the decoding of the uplink medium access control-control element.

[0070] In some example embodiments, the acknowledging downlink medium access control-control element may have a payload indicating the acknowledgment of decoding the uplink medium access control-control element by a bitmap.

[0071] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the terminal device to further perform: cancelling the trigger event if the number of transmission reaches the configured transmission number.

[0072] In some example embodiments, the number of transmission may be counted based on a number of initial transmission and retransmission of at least one transport block carrying the uplink medium access control-control element.

[0073] In some example embodiments, the number of transmission may be counted based on a number of initial transmission of at least one transport block carrying the uplink medium access control-control element.

[0074] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the terminal device to further perform: starting a timer configured by the network device for retriggering the trigger event, and retriggering the trigger event upon expiry of the timer if the number of transmission reaches the configured transmission number and without receiving, from the network device, a downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0075] In some example embodiments, the timer may be started at a last retransmission of the uplink medium access control-control element.

[0076] In some example embodiments, the timer may be started at an initial transmission of the uplink medium access control-control element.

[0077] In some example embodiments, the uplink medium access control-control element may be sent in the hybrid automatic repeat request process of mode B.

[0078] In some example embodiments, the transmission number may be common for a plurality of uplink medium access control-control elements.

[0079] In some example embodiments, the transmission number may be specific to the uplink medium access control-control element.

[0080] In an eighth aspect, a computer readable medium is disclosed. The computer readable medium may include instructions stored thereon for causing a network device to perform: configuring a transmission number for a terminal device to send an uplink medium access control-control element, wherein the uplink medium access control-control element may be sent to the network device in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

[0081] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the network device to further perform: transmitting, to the terminal device, a downlink medium access control-control element for acknowledging decoding of the uplink medium access control-control element if decoding the uplink medium access control-control element successfully.

[0082] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the network device to further perform: configuring the downlink medium access control-control element.

[0083] In some example embodiments, the computer readable medium may further include instructions stored thereon for causing the network device to further perform: configuring a timer for the terminal device to retrigger the trigger event, the trigger event being retriggered upon expiry of the timer if the number of transmission reaches the transmission number and the terminal device without receiving, from the network device, the downlink medium access control-control element acknowledging the decoding of the uplink medium access control-control element before the timer expires.

[0084] Other features and advantages of the example embodiments of the present disclosure will also be apparent from the following description of specific embodiments when read in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of example embodiments of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0085] Some example embodiments will now be described, by way of non-limiting examples, with reference to the accompanying drawings.

[0086] FIG. 1 shows an exemplary sequence diagram for MAC-CE transmission according to the example embodiments of the present disclosure.

[0087] FIG. 2 shows an exemplary flow diagram for MAC-CE transmission according to the example embodiments of the present disclosure.

[0088] FIG. 3 shows a flow chart illustrating an example method **300** for MAC-CE transmission according to the example embodiments of the present disclosure.

[0089] FIG. 4 shows a flow chart illustrating an example method **400** for MAC-CE transmission

according to the example embodiments of the present disclosure.

[0090] FIG. 5 shows a block diagram illustrating an example device **400** for MAC-CE transmission according to the example embodiments of the present disclosure.

[0091] FIG. 6 shows a block diagram illustrating an example device **600** for MAC-CE transmission according to the example embodiments of the present disclosure.

[0092] FIG. 7 shows a block diagram illustrating an example apparatus **700** for MAC-CE transmission according to the example embodiments of the present disclosure.

[0093] FIG. 8 shows a block diagram illustrating an example apparatus **800** for MAC-CE transmission according to the example embodiments of the present disclosure.

[0094] Throughout the drawings, same or similar reference numbers indicate same or similar elements. A repetitive description on the same elements would be omitted.

DETAILED DESCRIPTION

[0095] Herein below, some example embodiments are described in detail with reference to the accompanying drawings. The following description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known circuits, techniques and components are shown in block diagram form to avoid obscuring the described concepts and features.

[0096] Example embodiments of the present disclosure provide a solution that a MAC-CE can be transmitted on any UL resources despite of the HARQ mode so as to avoid transmission latency and meanwhile the reliability of correctly decoding a critical MAC-CE can be ensured.

[0097] FIG. 1 shows an exemplary sequence diagram for MAC-CE transmission according to the example embodiments of the present disclosure. Referring to the FIG. 1, a UE **110** may represent any terminal device in a wireless communication network. A network device **120** may function as a BS in the wireless communication network and serving the UE **110**. The wireless communication network may be e.g. a NTN, and the network device **120** may be located in a geosynchronous earth orbit (GEO) satellite or a non-GEO satellite, e.g. a medium earth orbit (MEO) satellite or a low earth orbit (LEO) satellite.

[0098] In an operation **125**, the network device **120** may configure a transmission number **130** for the UE **110** to send a UL MAC-CE. The transmission number **130** may be common for a plurality of UL MAC-CEs. Alternatively, the transmission number **130** may be specific for a set of UL MAC-CEs. Alternatively, the transmission number **130** may be specific for one UL MAC-CE, e.g. a UL MAC-CE **135**. The transmission number **130** may be a maximum number of transmission for the UE **110** to send the UL MAC-CE **135**, which may represent any UL MAC-CE. If a number of transmission is less than the configured transmission number **130**, the UE **110** may send the UL MAC-CE **135** in a HARQ process to the network device **120**.

[0099] The network device **120** may transmit the transmission number **130** to the UE **110**.

Obtaining the configured transmission number **130**, if a trigger event for the UL MAC-CE **135** occurs, in response to the trigger event, the UE **110** may in an operation **132** trigger the corresponding UL MAC-CE **135** and send, to the network device **120**, the UL MAC-CE **135** in a HARQ process if the number of transmission is less than the configured transmission number **130**. The transmission may be either an initial transmission of a transport block (TB) carrying the UL MAC-CE **135** or a retransmission of the TB carrying the UL MAC-CE **135**. The UL MAC-CE **135** may be multiplexed/transmitted in the UL grant scheduled HARQ process of either HARQ of mode A or HARQ of mode B.

[0100] In an embodiment, as an option, if the UL MAC-CE **135** is sent in the HARQ process of mode A, in an operation **170** the UE **110** may cancel the trigger event after e.g. first transmission of the UL MAC-CE **135** and refrain from further sending the UL MAC-CE **135** if the trigger event is cancelled. Because mode A may support reception of UL retransmission grant based on UL decoding result, the reliability of correctly decoding the MAC-CE **135** can be ensured to some

extent.

[0101] Alternatively, the UE **110** may refrain from canceling the trigger event even if the UL MAC-CE **135** is multiplexed/transmitted in the HARQ process of mode A. For example, the UE **110** may perform the following operations and/or cancel the trigger event based on other condition(s) regardless of HARQ mode A or HARQ mode B.

[0102] Alternatively, in some embodiments, the UE **110** may perform the following operations and/or cancel the trigger event based on other condition(s) if the UL MAC-CE **135** is sent in the HARQ process of mode B. HARQ mode B may support no UL retransmission and/or blind UL retransmission.

[0103] On the network side, if receiving the UL MAC-CE **135**, in an operation **145** the network device **120** may decode the UL MAC-CE **135**. If the network device **120** decodes the UL MAC-CE **135** successfully, the network device **120** may transmit, to the UE **110**, a downlink (DL) MAC-CE **155** for acknowledging the decoding of the UL MAC-CE **135**. The DL MAC-CE **155** may function as an acknowledgment (ACK) feedback for the UL MAC-CE **135**.

[0104] If receiving the DL MAC-CE **155**, in the operation **170**, the UE **110** may cancel the trigger event and refrain from further sending the UL MAC-CE **135** if the trigger event is cancelled. In an embodiment, the ACK DL MAC-CE **155** may have a logical channel identifier (LCID) or an extended logical channel identifier (eLCID) specific for the ACK of the decoding of the UL MAC-CE **135**. The LCID or the eLCID may be reserved for the UL MAC-CE **135**. In an option, the LCID or the eLCID may be in a subheader of the ACK DL MAC-CE **155**, and the ACK DL MAC-CE **155** may have no payload.

[0105] Alternatively, in an embodiment, the DL MAC-CE **155** may have a reserved bit specific for the ACK of the decoding of the UL MAC-CE **135**. For example, the DL MAC-CE **155** may be an existing MAC-CE having at least one reserved bit, and one reserved bit may be used for the ACK of the decoding of the UL MAC-CE **135**.

[0106] Alternatively or additionally, the ACK DL MAC-CE **155** may have a payload indicating the ACK of the decoding of the UL MAC-CE **135** by a bitmap. For example, the ACK DL MAC-CE **155** may have different bits for acknowledging the decoding of different UL MAC-CEs. For example, the third bit may be used for the ACK of the decoding of the UL MAC-CE **135**, and the fifth bit may be used for the ACK of the decoding of another UL MAC-CE **135**.

[0107] The mapping between the ACK DL MAC-CE **155** and the UL MAC-CE **135**, as well as the bitmap of the ACK DL MAC-CE **155** may be configured by the network device **120** in an operation **150** and be transmitted to the UE **110** via a radio resource control (RRC) signaling, alternatively or additionally, may be pre-specified in a specification/standard.

[0108] Alternatively or additionally, after the first transmission of the UL MAC-CE **135**, in an operation **140**, the UE **110** may count a number of transmission of the UL MAC-CE **135**, and may cancelling the trigger event if the number of transmission reaches the configured transmission number **130**.

[0109] In an embodiment, the UE **110** may count the number of transmission based on a number of initial transmission and retransmission of at least one TB carrying the UL MAC-CE **135**. For example, no matter the UL MAC-CE **135** is initially transmitted on a TB, retransmitted on the TB, or initially transmitted and retransmitted on another TB, the UE **110** will count as an increment of the number of transmission.

[0110] Alternatively, in an embodiment, the UE **110** may count the number of transmission based on the number of initial transmission of at least one TB carrying the UL MAC-CE **135**. For example, after the initial transmission on a TB, the retransmission on the TB will not be counted. The UE **110** will count as an increment of the number of transmission if the UL MAC-CE **135** is initially transmitted on another TB. In this case, the number of transmission may be the number of transmitted TBs carrying the UL MAC-CE **135**.

[0111] On the network side, in the operation **125**, the UE **110** may also configure a timer **160** for

the UE **110** to retrigger the trigger event. The timer **160** may relate to a duration allowing the network device **120** to confirm the decoding of the UL MAC-CE **135**. The configuration of the timer **160** may include a length of the timer. As an option, the configuration of the timer **160** may further include starting timing of the timer **160**. Alternatively, the starting timing of the timer **160** may be pre-specified in specification or standard. Receiving the configuration of the timer **160**, the UE **110** may use the timer **160** to retrigger the trigger event if the number of transmission reaches the transmission number **130** and the UE **110** has not received the ACK DL MAC-CE **155** before the timer **160** expires.

[0112] With the configured timer **160**, in an operation **165**, the UE **110** may start the timer **160**. In an embodiment, the UE **110** may start the timer **160** at a last retransmission of the UL MAC-CE **135**. In this case, the length of the timer **160** may be at least one round trip time (RTT) between the UE **110** and the network device **120**. Alternatively, in an embodiment, the UE **110** may start the timer **160** at an initial transmission of the UL MAC-CE **135**. In this case, the length of the timer **160** may be at least one RTT between the UE **110** and the network device **120** plus an estimated duration of retransmission(s) of the UL MAC-CE **135**.

[0113] In an operation **175**, the UE **110** may retrigger the trigger event upon expiry of the timer **160** if the number of transmission reaches the configured transmission number **130** and without receiving, from the network device **120**, the ACK DL MAC-CE **155** acknowledging the decoding of the UL MAC-CE **135** before the timer **160** expires. By retriggering the trigger event, the UE **110** may repeat the sending of the UL MAC-CE **135** such that the reliability of successfully decoding the UL MAC-CE **135** can be ensured

[0114] FIG. **2** shows an exemplary flow diagram for MAC-CE transmission according to the example embodiments of the present disclosure. Operations in the FIG. **2** will be briefly described if corresponding operations shown in the FIG. **1** have been described.

[0115] After a start of the flow, in an operation **210** the network device **120** may configure the maximum number of transmission, denoted as N_{max} , for a UL MAC-CE. The N_{max} may be e.g. the transmission number **130**, and the UL MAC-CE may be e.g. the UL MAC-CE **135**.

[0116] In an operation **215**, in response to the trigger event for the UL MAC-CE **135**, the UE **110** may trigger the corresponding UL MAC-CE **135** and set the number of transmission, denoted as N_{tx} , to be 0 and start the transmission of the UL MAC-CE **135** in a HARQ process.

[0117] As an option, in an operation **220**, if the UL MAC-CE **135** is multiplexed/sent in the HARQ process of mode A (“Yes” prong of the operation **220**), in an operation **225**, the UE **110** may cancel the trigger event and refrain from further sending the UL MAC-CE **135** if the trigger event is cancelled. If the UL MAC-CE **135** is multiplexed/sent in the HARQ process of mode B (“No” prong of the operation **220**), the UE **110** may continue the following operations.

[0118] Alternatively, the operation **220** may be omitted, and in this case the UE **110** may perform the flow despite the HARQ mode.

[0119] In an operation **230**, the UE **110** may count the N_{tx} . In an embodiment, the UE **110** may count the N_{tx} if a TB carrying the UL MAC-CE **135** is transmitted, regardless whether the UL MAC-CE **135** is initially transmitted on a TB, retransmitted on the TB or transmitted on other TB(s). If the HARQ process is set with a number of times for transmission, denoted as M , where the UL MAC-CE **135** will be transmitted M times, the N_{tx} may increase by M . For example, in a case where the UL MAC-CE **135** is multiplexed/transmitted in the HARQ process of mode B, if mode B supports blind retransmission, M may be equal to the number of times for the initial transmission and the following blind retransmission, and if mode B supports no UL retransmission, M will be one.

[0120] Alternatively in an embodiment, the UE **110** may count the N_{tx} as the number of initial transmissions of different TBs carrying the UL MAC-CE **135**. In this case the N_{tx} will increase by one regardless whether the retransmission is supported by the HARQ mode.

[0121] In an operation **235**, in a case where the N_{tx} has not reached the N_{max} (“Yes” prong of the

operation **235**), if in an operation **245** the UE **110** receives a DL ACK MAC-CE e.g. the ACK DL MAC-CE **155** acknowledging the decoding of the UL MAC-CE **135** (“Yes” prong of the operation **245**), in an operation **250**, the UE **110** may cancel the trigger event and refrain from further sending the UL MAC-CE **135** if the trigger event is cancelled. If in the operation **245** the UE **110** has not received the ACK DL MAC-CE **155** (“No” prong of the operation **245**), the UE **110** may continue the flow from the operation **220**.

[0122] In the operation **235**, if the Ntx reaches the Nmax (“No” prong of the operation **235**), in an operation **240**, the UE **110** may cancel the trigger event. After the operation **240** where the Ntx has reached the Nmax, if the UE **110** has not received the ACK DL MAC-CE **155** before e.g. the timer **160** expires, the UE **110** may retrigger the trigger event upon expiry of the timer **160**. In this case, the flow may repeat from the operation **215**.

[0123] It may be appreciated that the above embodiments may apply for any UL MAC-CE. Because there is no limitation on logical channel prioritization (LCP) on UL MAC-CE, a UL MAC-CE can be transmitted on any UL resources despite of the HARQ mode such that the UL MAC-CE transmission opportunities can be fully utilized and transmission latency can be avoided. Meanwhile the reliability of successfully decoding a UL MAC-CE can be ensured. The embodiments according to the present disclosure can result in more advantageous effects for a critical UL MAC-CE.

[0124] For example, for a UL MAC-CE reporting timing advance (TA), which may assist UL scheduling in the NTN and may be referred to as a TA report MAC-CE, if the TA report MAC-CE cannot be decoded by the network side successfully, the system performance will be severely impacted. The embodiments according to the present disclosure may be adaptive to the UL grant scheduled HARQ process despite the HARQ mode, and thus the reliability of successfully decoding a UL MAC-CE e.g. the TA report MAC-CE, can be achieved.

[0125] FIG. **3** shows a flow chart illustrating an example method **300** for MAC-CE transmission according to the example embodiments of the present disclosure. The example method **300** may be performed for example at a terminal device such as the UE **110**.

[0126] Referring to the FIG. **3**, the example method **200** may include an operation **310** of obtaining, a transmission number configured by a network device for sending a UL MAC-CE; and an operation **320** of sending, to the network device, the UL MAC-CE in a HARQ process if a number of transmission is less than the configured transmission number, in response to a trigger event for the UL MAC-CE.

[0127] Details of the operation **310** have been described in the above descriptions with respect to at least the transmission number **130** and the Nmax, and repetitive descriptions thereof are omitted here.

[0128] Details of the operation **320** have been described in the above descriptions with respect to at least the UL MAC-CE **135**, and repetitive descriptions thereof are omitted here.

[0129] In an embodiment, the example method **300** may further include an operation of cancelling the trigger event if the UL MAC-CE is sent in the HARQ process of mode A and refraining from further sending the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **170** and the operations **220** and **225**, and repetitive descriptions thereof are omitted here.

[0130] In an embodiment, the example method **300** may further include an operation of cancelling the trigger event if receiving, from the network device, a DL MAC-CE acknowledging decoding of the UL MAC-CE and refraining from further sending the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, the operation **170** and the operations **245** and **250**, and repetitive descriptions thereof are omitted here.

[0131] In an embodiment, the acknowledging DL MAC-CE may have a LCID or an eLCID specific for the ACK of the decoding of the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, and repetitive descriptions

thereof are omitted here.

[0132] In an embodiment, the LCID or the eLCID may be in a subheader of the acknowledging DL MAC-CE, and the acknowledging DL MAC-CE may be without a payload. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, and repetitive descriptions thereof are omitted here.

[0133] In an embodiment, the DL MAC-CE may have a reserved bit specific for the ACK of the decoding of the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, and repetitive descriptions thereof are omitted here.

[0134] In an embodiment, the acknowledging DL MAC-CE may have a payload indicating the ACK of decoding the UL MAC-CE by a bitmap. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, and repetitive descriptions thereof are omitted here.

[0135] In an embodiment, the example method **300** may further include an operation of cancelling the trigger event if the number of transmission reaches the configured transmission number. The more details have been described in the above descriptions with respect to at least the operation **170** and the operations **235** and **240**, and repetitive descriptions thereof are omitted here.

[0136] In an embodiment, the number of transmission may be counted based on a number of initial transmission and retransmission of at least one TB carrying the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **140** and the operation **230**, and repetitive descriptions thereof are omitted here.

[0137] In an embodiment, the number of transmission may be counted based on a number of initial transmission of at least one TB carrying the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **140** and the operation **230**, and repetitive descriptions thereof are omitted here.

[0138] In an embodiment, the example method **300** may further include an operation of starting a timer configured by the network device for retriggering the trigger event, the more details of which have been described in the above descriptions with respect to at least the operation **165**, and repetitive descriptions thereof are omitted here: and an operation of retriggering the trigger event upon expiry of the timer if the number of transmission reaches the configured transmission number and without receiving, from the network device, a DL MAC-CE acknowledging the decoding of the UL MAC-CE before the timer expires, the more details of which have been described in the above descriptions with respect to at least the operation **175** and the operation **255**, and repetitive descriptions thereof are omitted here.

[0139] In an embodiment, the timer may be started at a last retransmission of the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **165**, and repetitive descriptions thereof are omitted here.

[0140] In an embodiment, the timer may be started at an initial transmission of the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **165**, and repetitive descriptions thereof are omitted here.

[0141] In an embodiment, the UL MAC-CE may be sent in the HARQ process of mode B. The more details have been described in the above descriptions with respect to at least the operation **220**, and repetitive descriptions thereof are omitted here.

[0142] In an embodiment, the transmission number may be common for a plurality of UL MAC-CEs. The more details have been described in the above descriptions with respect to at least the transmission number **130**, and repetitive descriptions thereof are omitted here.

[0143] In an embodiment, the transmission number may be specific to the UL MAC-CE. The more details have been described in the above descriptions with respect to at least the transmission number **130**, and repetitive descriptions thereof are omitted here.

[0144] FIG. **4** shows a flow chart illustrating an example method **400** for MAC-CE transmission according to the example embodiments of the present disclosure. The example method **400** may be

performed for example at a network device such as the network device **120**.

[0145] Referring to the FIG. **4**, the example method **400** may include an operation **410** of configuring a transmission number for a terminal device to send a UL MAC-CE, wherein the UL MAC-CE may be sent to the network device in a HARQ process if a number of transmission is less than the configured transmission number, in response to a trigger event for the UL MAC-CE.

[0146] Details of the operation **410** have been described in the above descriptions with respect to at least the operation **125**, the transmission number **130**, the operation **210** and the Nmax, and repetitive descriptions thereof are omitted here.

[0147] In an embodiment, the example method **400** may further include an operation of transmitting, to the terminal device, a DL MAC-CE for acknowledging decoding of the UL MAC-CE if decoding the UL MAC-CE successfully. The more details have been described in the above descriptions with respect to at least the DL MAC-CE **155**, and repetitive descriptions thereof are omitted here.

[0148] In an embodiment, the example method **400** may further include an operation of configuring the DL MAC-CE. The more details have been described in the above descriptions with respect to at least the operation **150**, and repetitive descriptions thereof are omitted here.

[0149] In an embodiment, the example method **400** may further include an operation of configuring a timer for the terminal device to retrigger the trigger event, the trigger event being retriggered upon expiry of the timer if the number of transmission reaches the transmission number and the terminal device without receiving, from the network device, the DL MAC-CE acknowledging the decoding of the UL MAC-CE before the timer expires. The more details have been described in the above descriptions with respect to at least the timer **160** and the operation **170**, and the operation **255**, and repetitive descriptions thereof are omitted here.

[0150] FIG. **5** shows a block diagram illustrating an example device **400** for MAC-CE transmission according to the example embodiments of the present disclosure. The device, for example, may be at least part of a terminal device such as the UE **110** in the above examples.

[0151] As shown in the FIG. **5**, the example device **500** may include at least one processor **510** and at least one memory **520** that may include computer program code **530**. The at least one memory **520** and the computer program code **530** may be configured to, with the at least one processor **510**, cause the device **500** at least to perform the example method **300** described above.

[0152] In various example embodiments, the at least one processor **510** in the example device **500** may include, but not limited to, at least one hardware processor, including at least one microprocessor such as a central processing unit (CPU), a portion of at least one hardware processor, and any other suitable dedicated processor such as those developed based on for example Field Programmable Gate Array (FPGA) and Application Specific Integrated Circuit (ASIC). Further, the at least one processor **510** may also include at least one other circuitry or element not shown in the FIG. **5**.

[0153] In various example embodiments, the at least one memory **520** in the example device **500** may include at least one storage medium in various forms, such as a volatile memory and/or a non-volatile memory. The volatile memory may include, but not limited to, for example, a random-access memory (RAM), a cache, and so on. The non-volatile memory may include, but not limited to, for example, a read only memory (ROM), a hard disk, a flash memory, and so on. Further, the at least memory **520** may include, but are not limited to, an electric, a magnetic, an optical, an electromagnetic, an infrared, or a semiconductor system, apparatus, or device or any combination of the above.

[0154] Further, in various example embodiments, the example device **500** may also include at least one other circuitry, element, and interface, for example at least one I/O interface, at least one antenna element, and the like.

[0155] In various example embodiments, the circuitries, parts, elements, and interfaces in the example device **500**, including the at least one processor **510** and the at least one memory **520**, may

be coupled together via any suitable connections including, but not limited to, buses, crossbars, wiring and/or wireless lines, in any suitable ways, for example electrically, magnetically, optically, electromagnetically, and the like.

[0156] It is appreciated that the structure of the device on the side of the UE **110** is not limited to the above example device **500**.

[0157] FIG. **6** shows a block diagram illustrating an example device **600** for MAC-CE transmission according to the example embodiments of the present disclosure. The device, for example, may be at least part of a network device such as the network device **120** in the above examples.

[0158] As shown in the FIG. **6**, the example device **600** may include at least one processor **610** and at least one memory **620** that may include computer program code **630**. The at least one memory **620** and the computer program code **630** may be configured to, with the at least one processor **610**, cause the device **600** at least to perform the example method **400** described above.

[0159] In various example embodiments, the at least one processor **610** in the example device **600** may include, but not limited to, at least one hardware processor, including at least one microprocessor such as a central processing unit (CPU), a portion of at least one hardware processor, and any other suitable dedicated processor such as those developed based on for example Field Programmable Gate Array (FPGA) and Application Specific Integrated Circuit (ASIC). Further, the at least one processor **610** may also include at least one other circuitry or element not shown in the FIG. **6**.

[0160] In various example embodiments, the at least one memory **620** in the example device **600** may include at least one storage medium in various forms, such as a volatile memory and/or a non-volatile memory. The volatile memory may include, but not limited to, for example, a random-access memory (RAM), a cache, and so on. The non-volatile memory may include, but not limited to, for example, a read only memory (ROM), a hard disk, a flash memory, and so on. Further, the at least memory **620** may include, but are not limited to, an electric, a magnetic, an optical, an electromagnetic, an infrared, or a semiconductor system, apparatus, or device or any combination of the above.

[0161] Further, in various example embodiments, the example device **600** may also include at least one other circuitry, element, and interface, for example at least one I/O interface, at least one antenna element, and the like.

[0162] In various example embodiments, the circuitries, parts, elements, and interfaces in the example device **600**, including the at least one processor **610** and the at least one memory **620**, may be coupled together via any suitable connections including, but not limited to, buses, crossbars, wiring and/or wireless lines, in any suitable ways, for example electrically, magnetically, optically, electromagnetically, and the like.

[0163] It is appreciated that the structure of the device on the side of the network device **120** is not limited to the above example device **600**.

[0164] FIG. **7** shows a block diagram illustrating an example apparatus **700** for MAC-CE transmission according to the example embodiments of the present disclosure. The apparatus, for example, may be at least part of a terminal device such as the UE **110** in the above examples.

[0165] As shown in FIG. **7**, the example apparatus **700** may include means **710** for performing the operation **310** of the example method **300**, and means **720** for performing the operation **320** of the example method **400**. In one or more another example embodiments, at least one I/O interface, at least one antenna element, and the like may also be included in the example apparatus **700**.

[0166] In some example embodiments, examples of means in the example apparatus **700** may include circuitries. For example, an example of means **710** may include a circuitry configured to perform the operation **310** of the example method **300**, and an example of means **720** may include a circuitry configured to perform the operation **320** of the example method **300**. In some example embodiments, examples of means may also include software modules and any other suitable function entities.

[0167] FIG. 8 shows a block diagram illustrating an example apparatus **800** for MAC-CE transmission according to the example embodiments of the present disclosure. The apparatus, for example, may be at least part of a network device such as the network device **120** in the above examples.

[0168] As shown in FIG. 8, the example apparatus **800** may include means **810** for performing the operation **410** of the example method **400**. In one or more another example embodiments, at least one I/O interface, at least one antenna element, and the like may also be included in the example apparatus **800**.

[0169] In some example embodiments, examples of means in the example apparatus **800** may include circuitries. For example, an example of means **810** may include a circuitry configured to perform the operation **410** of the example method **400**. In some example embodiments, examples of means may also include software modules and any other suitable function entities.

[0170] The term “circuitry” throughout this disclosure may refer to one or more or all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); (b) combinations of hardware circuits and software, such as (as applicable) (i) a combination of analog and/or digital hardware circuit(s) with software/firmware and (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions); and (c) hardware circuit(s) and or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g., firmware) for operation, but the software may not be present when it is not needed for operation. This definition of circuitry applies to one or all uses of this term in this disclosure, including in any claims. As a further example, as used in this disclosure, the term circuitry also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.

[0171] Another example embodiment may relate to computer program codes or instructions which may cause an apparatus to perform at least respective methods described above. Another example embodiment may be related to a computer readable medium having such computer program codes or instructions stored thereon. In some embodiments, such a computer readable medium may include at least one storage medium in various forms such as a volatile memory and/or a non-volatile memory. The volatile memory may include, but not limited to, for example, a RAM, a cache, and so on. The non-volatile memory may include, but not limited to, a ROM, a hard disk, a flash memory, and so on. The non-volatile memory may also include, but are not limited to, an electric, a magnetic, an optical, an electromagnetic, an infrared, or a semiconductor system, apparatus, or device or any combination of the above.

[0172] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense: that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of

the items in the list, and any combination of the items in the list.

[0173] Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0174] As used herein, the term “determine/determining” (and grammatical variants thereof) can include, not least: calculating, computing, processing, deriving, measuring, investigating, looking up (for example, looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (for example, receiving information), accessing (for example, accessing data in a memory), obtaining and the like. Also, “determine/determining” can include resolving, selecting, choosing, establishing, and the like.

[0175] While some embodiments have been described, these embodiments have been presented by way of example, and are not intended to limit the scope of the disclosure. Indeed, the apparatus, methods, and systems described herein may be embodied in a variety of other forms: furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. For example, while blocks are presented in a given arrangement, alternative embodiments may perform similar functionalities with different components and/or circuit topologies, and some blocks may be deleted, moved, added, subdivided, combined, and/or modified. At least one of these blocks may be implemented in a variety of different ways. The order of these blocks may also be changed. Any suitable combination of the elements and actions of the some embodiments described above can be combined to provide further embodiments. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

[0176] Abbreviations used in the description and/or in the figures are defined as follows: [0177] ACK acknowledgment [0178] BS base station [0179] DL downlink [0180] eLCID extended logical channel identifier [0181] GEO geosynchronous earth orbit [0182] HARQ hybrid automatic repeat request [0183] LCID logical channel identifier [0184] LCP logical channel prioritization [0185] LEO low earth orbit [0186] MAC-CE medium access control-control element [0187] MEO medium earth orbit [0188] NTN non-terrestrial network [0189] RRC radio resource control [0190] RTT round trip time [0191] TA timing advance [0192] TB transport block [0193] UE user equipment [0194] UL uplink

Claims

1. A terminal device, comprising: at least one processor; and at least one memory comprising computer program code, the at least one memory and the computer program code being configured to, with the at least one processor, cause the terminal device to perform: obtaining, a transmission number configured by a network device for sending an uplink medium access control-control element; and sending, to the network device, the uplink medium access control-control element in a hybrid automatic repeat request process if a number of transmission is less than the configured transmission number, in response to a trigger event for the uplink medium access control-control element.

2.-44. (canceled)
